Deep learning

07/04/26

1) What is ANN?

→ It is a model inspired by the human brain consisting of neurons arranged in layers

Working

- Input layer receives data
- Hidden layers process data
- Output layer produces results

Mathematical Implementation

For one neuron:

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

$$a = f(z)$$

where w = weights, x = input, b = bias, $f(z)$ = act function

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2) What is CNN?

→ Used for image processing and extracts features using filters

Working

- Convolution layer
- Activation (ReLU)
- Pooling layer
- Fully connected layer

Mathematical Implementation

Convolution operation:

$$\text{output}(i,j) = \sum x(i+k, j+l) \cdot w(k,l)$$

3) What is RNN?

→ used for sequential data like text or time series

Working

- Uses previous output as input
- maintains memory

mathematical implementation

$$h_t = f(Wx_t + Uh_{t-1})$$

$$y_t = Wh_t$$

4) What is LSTM?

→ A special RNN that solves long term dependency problem

Working

Uses gates:

- Forget gate
- Input gate
- Output gate

Mathematical Implementation

Forget gate: $f_t = \sigma(W_f [h_{t-1}, x_t])$

Input gate: $i_t = \sigma(W_i [h_{t-1}, x_t])$

Cell state: $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$

Output: $h_t = O_t * \tanh(C_t)$

5) What is GRU?

→ Simplified version of LSTM

Working

- Update gate
- Reset gate

Mathematical Implementation

$$z_t = \sigma(W_z x_t + U_z h_{t-1})$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1})$$

$$h_t = (1 - z_t) h_{t-1} + z_t \tilde{h}_t$$

6) What is GAN?

→ has two models

- Generator
- Discriminator

GAN working

- Generator creates fake data
- Discriminator checks real vs fake
- Both compete

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$$\min_G \max_D V(D,G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$

7) What is VAE?

→ Used to generate new data by learning probability distribution

working

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$$L = \text{Reconstruction} + \text{KL Divergence}$$

$$KL(q(z|x) \| p(z))$$